

Global Climate Events Economic Impact

Project Documentation

Team Members

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 - Mohamed Emad (Team Ledar)
 - Shaimaa Abdelrahman
 - Amal Ayman
 - Basant Abdelhakim
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1. Project Planning & Management

1.1 Project Proposal

Project Overview

The Global Climate Events Economic Impact project focuses on analyzing worldwide climate disasters and their economic and human impact between 2020 and 2025. The project aims to transform raw climate event data into meaningful insights through data cleaning, analysis, visualization, and dashboard development using multiple analytical tools.

The project was implemented using:

- Microsoft Excel
- SQL
- Python
- Power BI
- Tableau

Each tool was used independently to perform the complete analysis workflow including data cleaning, analysis, KPI generation, and dashboard creation.

Project Objectives

Project Scope

The project covers:

- Data cleaning and preprocessing.
- Exploratory data analysis.
- KPI generation.
- Statistical and trend analysis.
- Dashboard development.
- Visualization and reporting.

The project does not include:

- Machine learning models.
- Real-time data streaming.
- Web or mobile application deployment.

1.2 Project Plan

1.3 Task Assignment & Roles

Team Member	Responsibilities
Ahmed Bahaa	SQL analysis, Python analysis
Mohamed Emad	SQL analysis, Python analysis
Shaimaa Abdelrahman	Excel dashboard, Power BI, Tableau
Amal Ayman	Excel dashboard, Power BI, Tableau
Basant Abdelhakim	Excel dashboard, Power BI, Tableau

1.4 Risk Assessment & Mitigation Plan

Risk	Impact	Mitigation
Missing values in dataset	Inaccurate analysis	Data cleaning and validation
Incorrect data types	Calculation errors	Data preprocessing and conversion
Dashboard performance issues	Slow visualization	Optimization of calculations
Team coordination issues	Delayed progress	Task distribution and regular follow-up
Data inconsistency across tools	Different outputs	Standardized cleaning process

1.5 KPIs (Key Performance Indicators)

The following KPIs were used in dashboards and analysis:

- Total Events
- Total Deaths
- Total Economic Impact
- Average Severity
- Average Response Time

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Suggested Improvements

- Add more geospatial analysis.
- Improve dashboard interactivity.
- Expand statistical analysis.
- Add predictive analysis in future versions.

Final Grading Criteria

Criteria	Weight
Documentation	20%
Data Analysis	30%
Dashboard Design	20%
Presentation	15%
Team Collaboration	15%

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Needs
Analysts	Accurate climate insights
Decision Makers	Visual dashboards and KPIs
Researchers	Trend and impact analysis
Emergency Management Teams	Risk and response evaluation

3.2 User Stories & Use Cases

User Story Example

As a decision maker, I want to analyze climate disaster trends so that I can improve disaster preparedness strategies.

Use Cases

- View total climate events.
 - Analyze economic impact by country.
 - Compare event types.
 - Analyze seasonal trends.
 - Evaluate emergency response efficiency.
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3.3 Functional Requirements

- Import and clean dataset.
 - Generate KPIs.
 - Perform aggregations and statistical analysis.
 - Create interactive dashboards.
 - Visualize trends and comparisons.
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3.4 Non-Functional Requirements

Requirement	Description
Performance	Fast dashboard loading
Reliability	Accurate calculations
Usability	Easy dashboard navigation
Scalability	Ability to handle larger datasets
Maintainability	Organized and readable code

4. System Analysis & Design

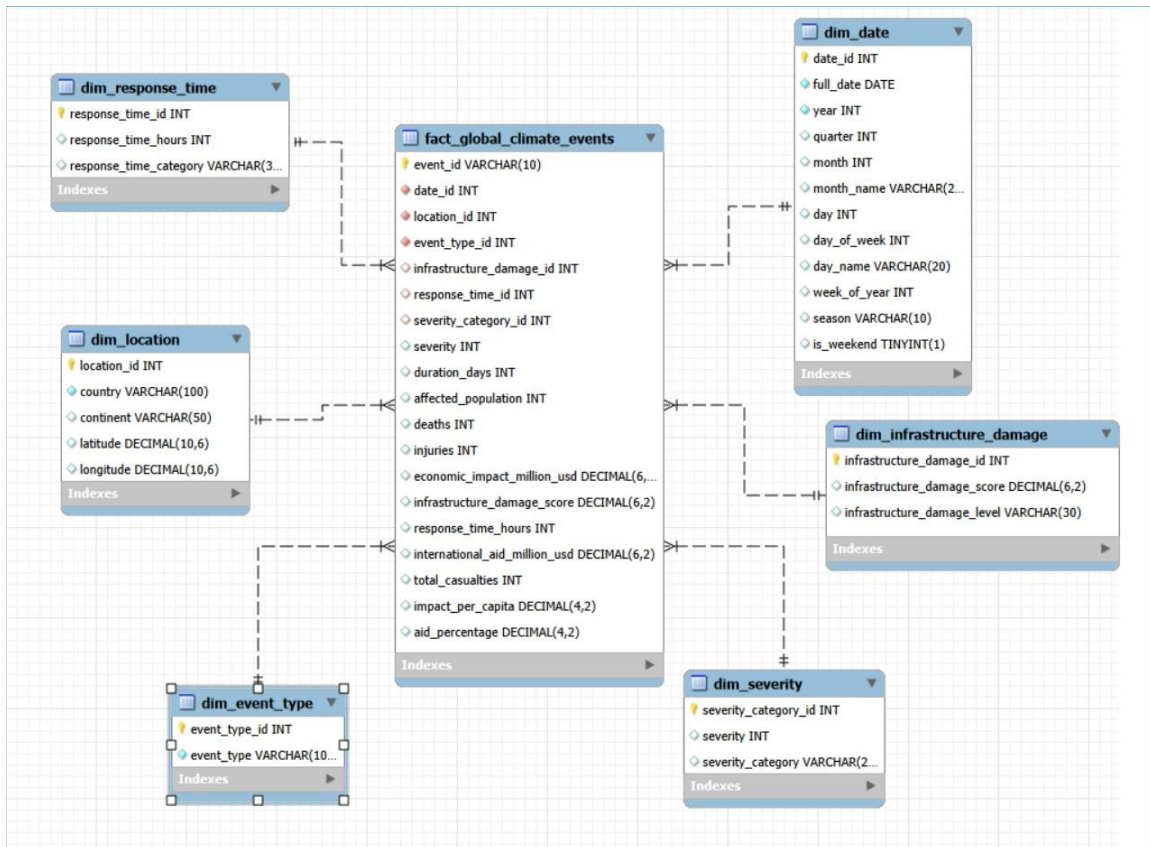
4.1 Problem Statement & Objectives

Problem Statement

Climate disasters are increasing globally, causing major economic losses and human casualties. Analyzing this data manually is difficult due to the large number of events and variables.

Project Goals

- Simplify climate data analysis.
 - Provide meaningful insights.
 - Support data-driven decision making.
 - Visualize climate disaster patterns.
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Software Architecture

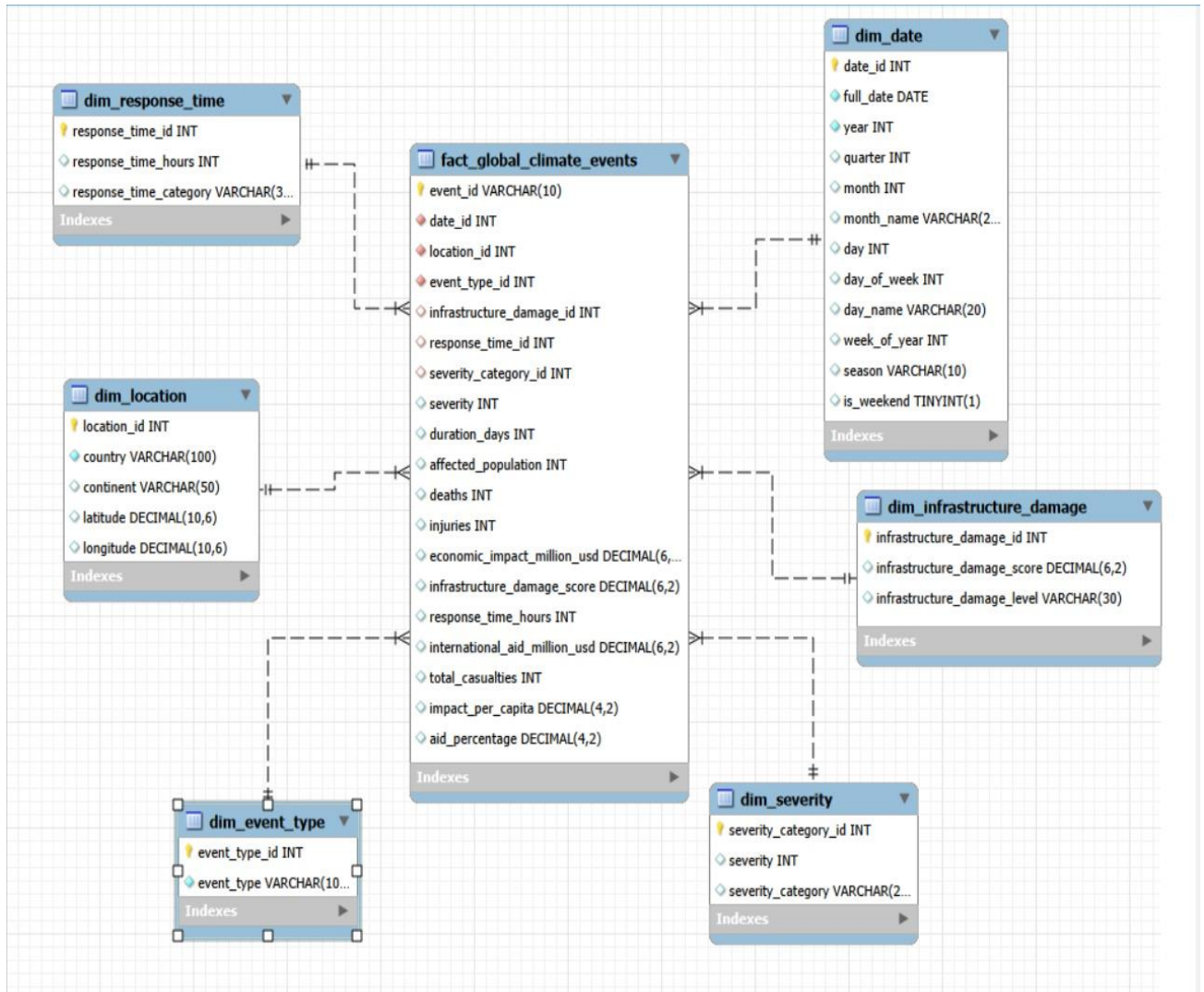
The project follows a data analytics workflow architecture:

1. Data Collection
2. Data Cleaning
3. Data Analysis
4. Visualization & Dashboarding

5. Reporting & Documentation

4.2 Database Design & Data Modeling

ER Diagram



4.5 System Deployment & Integration

Technology Stack

Technology	Usage
Excel	Data cleaning and dashboarding
SQL	Querying and aggregations
Python	Data analysis and visualization
Power BI	Interactive dashboard creation
Tableau	Data visualization and dashboards

4.6 Additional Deliverables

Testing & Validation

- Validation of calculations across tools.
 - Dashboard KPI consistency checks.
 - Verification of aggregations and statistical outputs.
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5. Implementation (Source Code & Execution)

5.1 Source Code

Structured & Well-Commented Code

The project code was organized into:

- Data cleaning steps.
 - Exploratory analysis.
 - Aggregations and grouping.
 - KPI calculations.
 - Visualization sections.
 - Dashboard preparation.
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Coding Standards & Naming Conventions

- Meaningful variable names.
 - Consistent formatting.
 - Readable code structure.
 - Reusable analysis sections.
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Modular Code & Reusability

The analysis process was divided into reusable sections including:

- Cleaning functions.
 - Aggregation logic.
 - KPI generation.
 - Visualization workflows.
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Security & Error Handling

- Missing value handling.
 - Duplicate removal.
 - Data type validation.
 - Aggregation verification.
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5.2 Version Control & Collaboration

The project was developed collaboratively by the team with shared task distribution and continuous review.

5.3 Deployment & Execution

README Contents

Installation Steps

1. Install Python and required libraries.
 2. Open SQL environment.
 3. Load dataset.
 4. Run analysis scripts.
 5. Open dashboard files.
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System Requirements

- Python 3.x
 - Jupyter Notebook
 - Microsoft Excel
 - Power BI Desktop
 - Tableau Public/Desktop
 - SQL Environment
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Execution Guide

- Run SQL scripts for querying.
 - Execute Python notebook cells.
 - Open dashboard files in Power BI and Tableau.
 - Use Excel dashboard for KPI interaction.
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6. Testing & Quality Assurance

Test Cases & Test Plan

Test Case	Expected Result
Missing value handling	Clean dataset without null issues
KPI calculations	Accurate totals and averages
Dashboard filters	Correct visual updates
SQL aggregations	Accurate grouped outputs

Bug Reports

Issue	Resolution
Incorrect data types	Converted columns properly
Duplicate rows	Removed duplicates
Visualization formatting issues	Improved chart formatting

7. Final Presentation & Reports

User Manual

Dashboard Usage

- Navigate between dashboard pages.
 - Use filters for country and event type analysis.
 - Analyze KPIs and charts.
 - Compare trends across years and seasons.
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Technical Documentation

The project includes:

- SQL analysis scripts.
 - Python notebook.
 - Dashboard files.
 - Database schema.
 - Documentation report.
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Project Presentation

The final presentation summarizes:

- Project objectives.
 - Dataset overview.
 - Data cleaning process.
 - Analysis workflow.
 - Dashboard demonstrations.
 - Key insights and recommendations.
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Key Insights

- Climate events remained consistently high across years.
 - Summer recorded the highest event severity and casualties.
 - Autumn generated the highest total economic impact.
 - Some event types show strong seasonal patterns.
 - Faster emergency response may reduce casualties.
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Recommendations

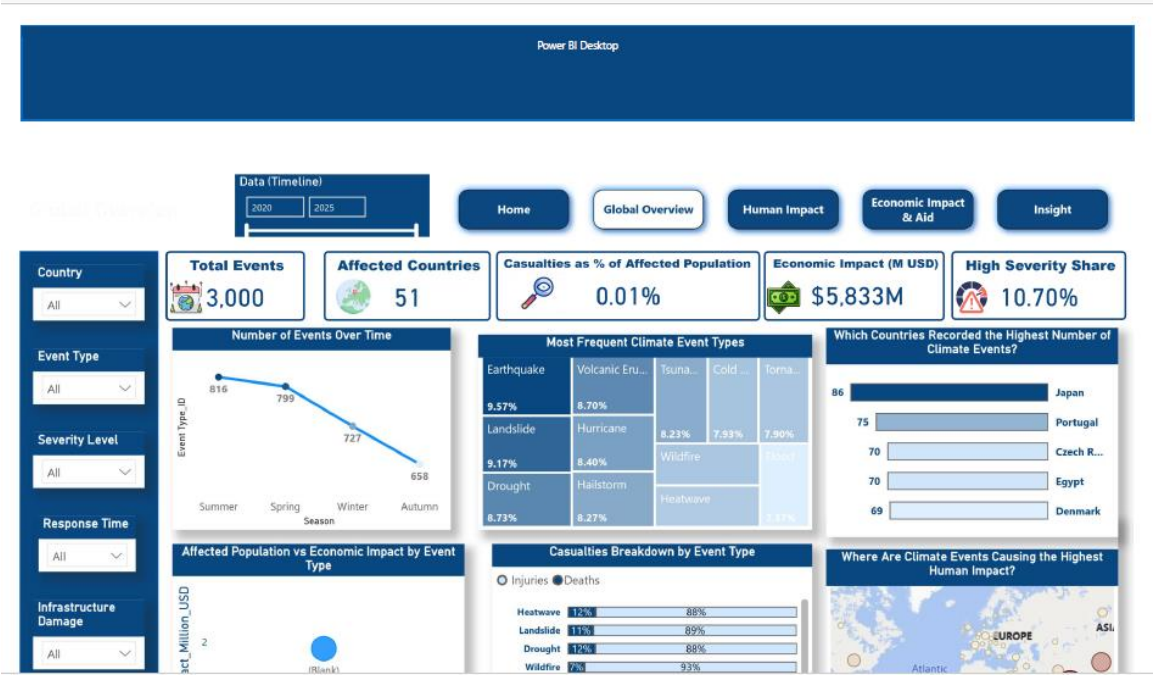
- Improve emergency response systems.
 - Increase investment in climate-resilient infrastructure.
 - Enhance disaster preparedness strategies.
 - Strengthen seasonal risk management plans.
 - Improve international cooperation during climate disasters.
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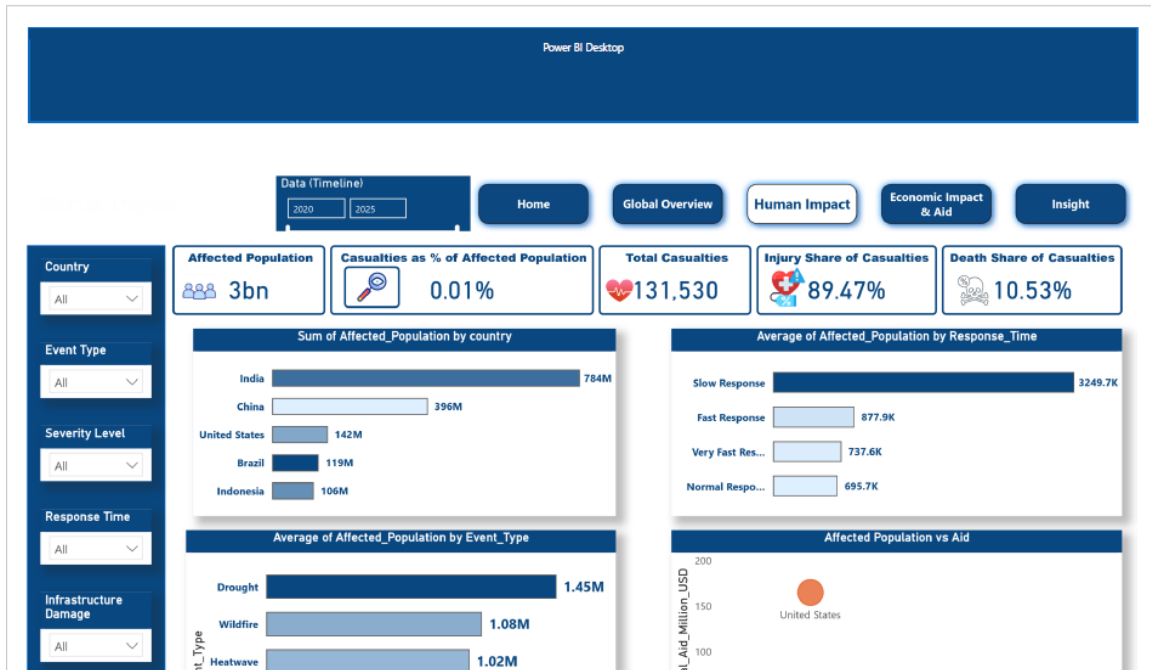
Appendix

Dataset Source

Dataset obtained from Kaggle.

Dashboard Screenshots





Python Analysis Screenshots

[Place Python Visualizations Here]





GLOBAL CLIMATE EVENTS - ECONOMIC IMPACT DASHBOARD

DATA ANALYST SPECIALIST TRACK - FINAL PROJECT



2020 2025

All Continents

All Event Types

All Severities

All Seasons

Total Events

2,930

Total Deaths

13,484

Economic Impact

\$5.8B

Avg Severity (1/10)

4

Avg Response Time

11 hrs

Events with Aid

0.1%

Temporal Patterns

Geographic Impact

Human Toll

Response & Aid

Strategic Analytics

HUMAN TOLL - PAGE 3 OF 5

Top 10 Deadliest Disaster Types



Top 5 Deadliest Events per Year (Cumulative)



Aid Impact: Deaths & Economic Loss



Disaster Family Lethality Rate (Deaths / Total Casualties)



Catastrophic Events (Severity 7+): Deaths by Type



GLOBAL CLIMATE EVENTS - ECONOMIC IMPACT DASHBOARD

DATA ANALYST SPECIALIST TRACK - FINAL PROJECT



2020 2025

All Continents

All Event Types

All Severities

All Seasons

Total Events
2,930

Total Deaths
13,484

Economic Impact
\$5.8B

Avg Severity (1/10)
4

Avg Response Time
11 hrs

Events with Aid
0.1%

Temporal Patterns

Geographic Impact

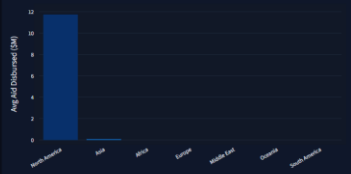
Human Toll

Response & Aid

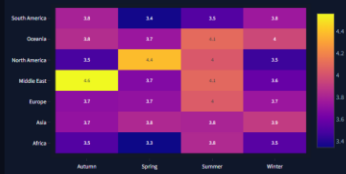
Strategic Analytics

Geographic Impact - PAGE 2 OF 5

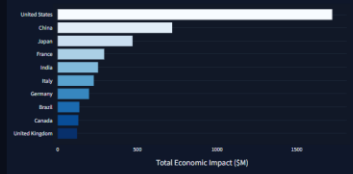
Funding Gap: Avg Aid for High-Severity Events (~6)



Avg Disaster Severity: Continent x Season



Top 10 Most Financially Impacted Countries



Deadliest Disaster Types by Continent



Yearly Economic Impact by Continent

